

BIS 629 Guest Lecture

Cluster Randomized Trials with Constrained Randomization: Design and Analysis

Fan Li

Center for Methods in Implementation and Prevention Science (CMIPS)
Department of Biostatistics
Yale School of Public Health

Cluster (group) randomized trials

- ▶ Randomization at the cluster level (clinics, hospitals, etc.)
 - ▶ Intervention delivered at the cluster level
 - ▶ Outcomes often measured at the individual level
- ▶ We focus on parallel-arm designs
 - ▶ Intervention implemented simultaneously in half of the clusters
- ▶ A practical limitation in designing a parallel CRT is the ability to enlist many clusters
 - ▶ $\geq 50\%$ of approx. 300 published CRTs randomize ≤ 21 or fewer clusters (Ivers et al. 2011)
- ▶ **Chance imbalance** likely to occur after simple randomization (an example to follow)

Reminder/Recall Immunization CRT

(Dickinson et al. 2015)

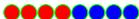
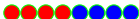
- ▶ Consider the reminder/recall (R/R) immunization study
 - ▶ 2-arm parallel CRT with 16 counties (clusters)
 - ▶ to increase immunization rate in children 19-35 months
 - ▶ a population-based R/R approach (Treatment Arm)
 - ▶ a practice-based R/R approach (Control Arm)
 - ▶ binary response variable, immunization status for children
- ▶ Location known for all clusters (8 rural & 8 urban)

Ideal scenario

- Symbolic representation

Location	# of counties	Symbols
Rural	8	
Urban	8	

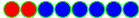
- Assign 8 counties to each arm
- Often times, we wish to achieve some sort of “balance” after randomization

Arm	# of rural/urban counties	Symbols
Trt	4/4	
Ctr	4/4	

- Same number of urban (or rural) counties/arm $\Rightarrow$ balance

Chance imbalance

- ▶ Random allocation of 16 counties to two arms does not guarantee “balance”
- ▶ Under *simple randomization*, we may end up getting

Arm	# of rural/urban counties	Symbols
Trt	2/6	
Ctr	6/2	

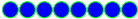
- ▶ With a few clusters, the probability of getting an “imbalanced” random allocation is not negligible ($\approx 1/8$)
- ▶ Chance imbalance becomes a bigger issue with ≥ 1 baseline covariate

Why need baseline balance at all?

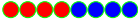
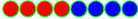
- ▶ Chance imbalance leads to
 - ▶ poor internal validity
 - ▶ reduced study power/precision of estimates (issue magnified by small sample size)
- ▶ Therefore need **design-based adjustment** of covariates to reduce chance imbalance
- ▶ Design-based solution is possible since
 - ▶ all clusters are identified **prior to** randomization (cluster characteristics available)
 - ▶ in contrast to individually randomized trials with continuous enrollment

Stratification

- ▶ With only one cluster-level covariate (C)
 - ▶ can create distinct strata of clusters based on baseline covariates
- ▶ Stratified randomization

	Location	Symbols	Randomization
Stratum 1	rural		1 : 1 to two arms
Stratum 2	urban		1 : 1 to two arms

- ▶ Balance is maintained within each stratum defined by location

Arm	# of rural/urban counties	Symbols
Trt	4/4	
Ctr	4/4	

- ▶ What if baseline covariate C is continuous?

Baseline characteristics in R/R immunization study

- ▶ Various types of covariates, most of which are continuous
- ▶ ① location (rural/urban); ② % children with immunization record; ③ % up-to-date at baseline; ④ % Hispanic; ⑤ average income category, and so on

clustername	location	inciis	uptodateonimmunizations	hispanic	incomecat
1	Rural	94	37	44	Low
2	Rural	85	39	23	High
3	Rural	85	42	12	Low
4	Rural	93	39	18	High
5	Rural	82	31	6	High
6	Rural	80	27	15	Med
7	Rural	94	49	38	Low
8	Rural	100	37	39	Low
9	Urban	93	51	35	Med
10	Urban	89	51	17	Med
11	Urban	83	54	7	High
12	Urban	70	29	13	Med
13	Urban	93	50	13	High
14	Urban	85	36	10	Med
15	Urban	82	38	39	Low
16	Urban	84	43	28	Med

Stratification with two covariates?

- ▶ Create distinct strata of clusters based on baseline covariates
 - ▶ continuous variables will be categorized (e.g. *high* versus *low*)

	Location	Avg income	# of counties	Randomization
Stratum 1	rural	low	○	1 : 1 to two arms?
Stratum 2	rural	medium	○○○	1 : 1 to two arms?
Stratum 3	rural	high	○○○○	1 : 1 to two arms?
Stratum 4	urban	low	none	none
Stratum 5	urban	medium	○○○○○	1 : 1 to two arms?
Stratum 6	urban	high	○○○	1 : 1 to two arms?

- ▶ Con: incomplete filling of strata with ↑ number of strata
 - ▶ unavoidable with a number of baseline covariates (R/R study)
 - ▶ sensitive to cutoff used in categorization
 - ▶ same drawback in individual RCTs

Pair matching

- ▶ Creating good matches $\Rightarrow$ an effective mechanism to create comparable groups
- ▶ Suppose location variable is of good prognostic values (the matching variable), can create eight pairs of clusters

	rural/urban counties	Symbols	Trt	Ctr
Pair 1	2/0			
Pair 2	2/0			
Pair 3	2/0			
Pair 4	2/0			
Pair 5	0/2			
Pair 6	0/2			
Pair 7	0/2			
Pair 8	0/2			

Pair matching

- ▶ Matching with multiple covariates relies on a multivariate distance metric
- ▶ Advantage
 - ▶ allows for an efficient and simple **design-based estimator** (Imai, King and Nall, Stat Sci 2009)
- ▶ Disadvantages (Klar and Donner, Stat Med 1997)
 - ▶ loss of follow-up from one cluster removes its matches
 - ▶ difficult to properly calculate the intraclass correlation coefficient (ICC)
 - ▶ Hard to find good matches with small number of clusters, and ineffective matching – “**break the matches**”?

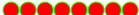
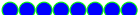
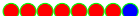
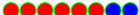
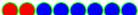
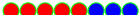
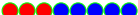
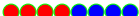
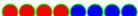
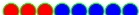
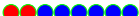
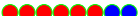
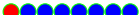
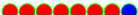
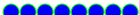
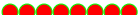
Constrained randomization (CR)

(Raab and Butcher, 2001; Moulton, 2004)

- ▶ General idea
 - ▶ Characterize the **simple randomization space** containing all possible allocation schemes
 - ▶ Define and assess “balance” for each possible allocation scheme
 - ▶ Randomize only within a constrained space with “balanced” allocation schemes
- ▶ Advantages
 - ▶ remove **bad-luck** randomization
 - ▶ accomodate a number of, and all types of covariates
 - ▶ does not complicate ICC calculation
- ▶ Equivalent to **re-randomization**
 - ▶ only accept a randomization if balance falls under a balance threshold

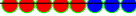
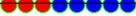
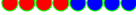
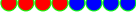
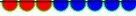
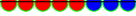
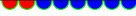
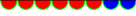
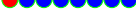
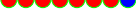
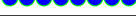
Schematic illustration of constrained randomization

- ▶ R/R study with $n = 16$ clusters and 8 clusters/arm
- ▶ Simple randomization space: 12,870 allocation schemes
- ▶ 9 allocation types of 8 rural ($C = 1$) & 8 urban ($C = 1$) clusters
- ▶ Balance score by a $l1$ balance metric: $|\bar{C}_T - \bar{C}_C|$

# Rural in Arms	Treatment	Control	# of schemes	Balance
8/0			1	1.00
7/1			64	0.75
6/2			784	0.50
5/3			3136	0.25
4/4			4900	0.00
3/5			3136	0.25
2/6			784	0.50
1/7			64	0.75
0/8			1	1.00

Schematic illustration of constrained randomization

- ▶ Constrain to 4,900/12,870 allocations with most balance
 - ▶ Balance score = 0
 - ▶ 4 rural & 4 urban clusters/arm
- ▶ Randomize 16 clusters within the constrained subset of 4,900

Treatment	Control	# of schemes	Balance
		1	1.00
		64	0.75
		784	0.50
		3136	0.25
		4900	0.00
		3136	0.25
		784	0.50
		64	0.75
		1	1.00

- ▶ With one binary covariate, CR generalizes stratification

Formalizing Constrained Randomization

- ▶ Step 1: Specify important baseline cluster-level covariates $\mathcal{C}$
- ▶ Step 2: Generate allocation schemes
 - ▶ Either enumerate all schemes (e.g. if $n \leq 18$)
 - ▶ Or simulate many schemes (e.g. 50,000) & remove duplicates
- ▶ Step 3: Select a **constrained randomization space** with sufficiently-balanced allocations according to **balance metric**
- ▶ Step 4: Randomly sample 1 scheme from constrained randomization space

Question: what is an equivalent way for **re-randomization**? $\Rightarrow$ recognize the **duality** between size of the randomization space and the balance threshold

Design Parameter 1: Balance metrics

- ▶ Consider a parallel-arm design with K clusters
 - ▶ n_k : cluster size
 - ▶ X_k : intervention status for cluster k (assume binary for now)
 - ▶ $\mathbf{C}_k = (C_{k,1}, \dots, C_{k,p_1})'$: cluster-level covariates
 - ▶ $\mathbf{S}_{kj} = (S_{kj,1}, \dots, S_{kj,p_2})'$: subject-level covariates
 - ▶ $\mathbf{Z}_{kj} = (\mathbf{C}'_k, \mathbf{S}'_{kj})'$: collection of all covariates
 - ▶ y_{kj} : outcome for subject j in cluster j
- ▶ At the design stage, often only $\mathbf{C}_k$ is available – either **pure** cluster-level covariates, or **summary statistics** of $\mathbf{S}_{kj}$ from historical data
- ▶ Could consider balance metric as any sensible distance function in $\mathbf{C}_k$ between arms

Design Parameter 1: Balance metrics

- ▶ Commonly used l_2 balance metric (Raab and Butcher, Stat Med 2001)

- ▶ define $\bar{Z}_l^{(1)}$ and $\bar{Z}_l^{(0)}$ as the average of l th covariate among the treated and control group, then use **Euclidean distance**

$$B_{(l_2)} = \sum_l \omega_l \left(\bar{Z}_l^{(1)} - \bar{Z}_l^{(0)} \right)^2$$

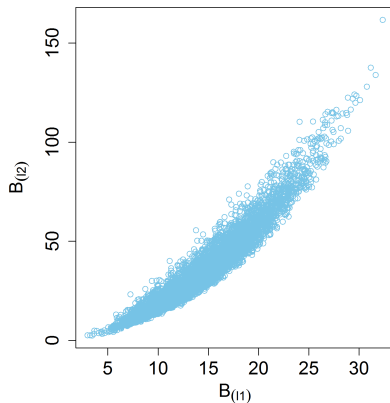
- ▶ can specify values of ω_l to up-weight or down-weight certain baseline covariates to reflect different priorities
 - ▶ default choice of ω_l as inverse variance of the group means for l th covariate
 - ▶ dummy variables can be used for multi-categorical covariate
 - ▶ invariant to **affine transformation** of covariates with the default weight choice
- ▶ The l_1 counterpart (Li et al., Stat Med 2017):
$$B_{(l_1)} = \sum_l \tilde{\omega}_l \left| \bar{Z}_l^{(1)} - \bar{Z}_l^{(0)} \right|$$

Design Parameter 1: Balance metrics

- ▶ Other balance metrics
 - ▶ maximum of absolute standardized difference as in observational studies
 - ▶ Mahalanobis distance (Morgan and Rubin, AOS 2015; Zhou et al., 2022)
 - ▶ ...
- ▶ Empirical studies indicate that different balance metrics tend to perform **similar**
 - ▶ relative ranking of randomization scheme matters more than absolute balance score
- ▶ Side note: with default weight, there exists **one-step matrix formulas** to compute all $l1$ and $l2$ balance scores given any randomization space without looping (Appendix B in Li et al., Stat Med 2017)

R/R Immunization Study: Two balance metrics

- ▶ Balance all 9 baseline covariates
- ▶ $l1$ and $l2$ metrics very similar: can use either one for constrained randomization



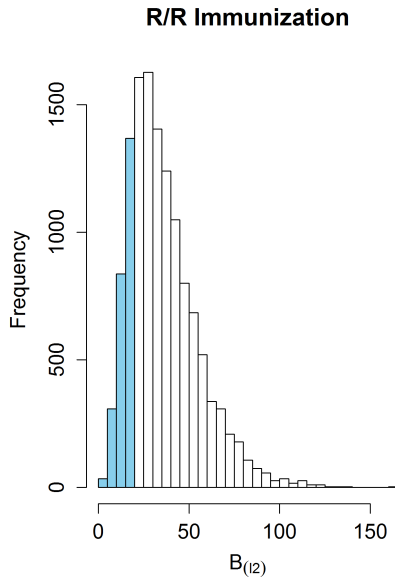
- ▶ Spearman rank correlation: $\lambda = 0.97$

Design Parameter 2: Size of Randomization Space

- ▶ Once balance metric specified, obtain $\{(I_r, B_r) : r = 1, \dots, R\}$, with I_r, B_r denoting the r th allocation scheme and its balance score, and R representing the maximum number of distinct schemes under **complete enumeration** or **random simulation**
- ▶ Define $q \in (0, 1]$ corresponds to constrained randomization space $\mathbb{C}_q = \{I_r : B_r \leq F_B^{-1}(q), r = 1, \dots, R^*\}$
 - ▶ $F_B^{-1}(q)$: quantile function (empirical CDF) of the balance score
 - ▶ R^* : # of distinct schemes retained in the constrained space
 - ▶ $q = 1$: simple randomization (SR)
- ▶ **q small but should not be overly small**
 - ▶ Risk **deterministic allocation**
 - ▶ May prohibit permutation test for a fixed α (more later)
- ▶ $q = 10\%$ works well in simulations
 - ▶ Power $\uparrow$ as $q \downarrow$ by balancing predictive covariates
 - ▶ Relationship not monotone, power may even $\downarrow$ if $q \rightarrow 0$

R/R Immunization Study

- ▶ Balance all 9 baseline covariates
- ▶ $\binom{16}{8} = 12,870$ possible allocation schemes with balanced assignment
- ▶ Example: constrained randomization space 10% of simple randomization space



Overly Constrained Randomization

- ▶ What is an optimal q ?
 - ▶ currently unclear, but from an efficiency point of view, q^{opt} can depend on **the estimator** of treatment effect & **data generating process**
 - ▶ efficiency of common model-based estimator seems to be quadratic in q (only an empirical conjecture)
- ▶ Beyond efficiency, if q is too small (say $q \approx 0$), then knowing the treatment status of cluster k may completely reveal the treatment status of cluster k'
 - ▶ certain pairs may always be allocated to the same arm (**cluster coincidence**)
 - ▶ certain pairs may never be allocated to the same arm (**cluster separation**)
 - ▶ loss of randomization validity

Check for Randomization Validity

- ▶ Selection of q also informed by randomization validity
- ▶ Obtain summary statistics on cluster pairs that always or never appear together in the same arm under $\mathbb{C}_q$
 - ▶ available in SAS macro (Greene, JSS 2017) and R package `cvcrand` (Yu et al., R Journal 2019)
- ▶ example with the R/R study

```
> Design_result$cluster_coin_des
```

	Mean	Std Dev	Minimum	25th Pctl	Median	75th Pctl	Maximum
samecount	600.600	88.807	368.000	551.750	603.000	648.500	804.000
samefrac	0.467	0.069	0.286	0.429	0.469	0.504	0.625
diffcount	686.400	88.807	483.000	638.500	684.000	735.250	919.000
difffrac	0.533	0.069	0.375	0.496	0.531	0.571	0.714

- ▶ the cluster pairs has a 47% chance on average to appear in the same arm (close to 50%)
- ▶ there is a cluster pair that will appear in the same arm for only about 29% of the times
- ▶ Greene (2017) considered 25% and 75%, as thresholds for loss of validity.

Example: R/R Immunization study with 3 covariates

- ▶ Allocate 16 clusters (8/arm) + balance on 3 covariates
- ▶ Use $B_{(I2)}$ metric + constrain at $q = 10\%$ of simple randomization (SR) space
- ▶ Compare mean covariate levels between arms under “best” balance, at “boundary” of 10% CR space & under “worst” balance (i.e. at worst SR allocation)

Covariate - Mean	“Best”		“CR Boundary”		“Worst”	
	Trt	Ctr	Trt	Ctr	Trt	Ctr
# of urban county	4	4	4	4	0	8
% hispanic	22.4	22.3	19.9	24.8	24.4	20.3
% up-to-date at baseline	40.8	40.9	40.3	41.4	37.6	44.0
balance score	$B_{(I2)} = 0.005$		$B_{(I2)} = 2.58$		$B_{(I2)} = 71.08$	

Application of CR to the R/R study with all 9 covariates

(Li et al., Stat Med 2016)

- Balance all of 9 baseline covariates
- “best” and “worst” allocation schemes under CR with $q = 0.1$

Covariate - Mean	“Best”		“CR Boundary”	
	Trt	Ctr	Trt	Ctr
# of urban county	4	4	4	4
% in CHS	87.4	87.0	87.6	86.6
# of children	4172	4221	4068	4325
% up-to-date at baseline	41.4	40.3	42.1	39.5
% African American	3.3	2.5	3.5	2.3
% Hispanic	21.9	22.8	18.3	26.4
Average income (\$1000/yr)	54.8	52.2	51.3	55.7
PM-to-FM ratio	0.26	0.30	0.23	0.33
# CHCs	4.4	4.4	4.5	4.3
balance score	$B_{(I2)} = 2.5$		$B_{(I2)} = 15.4$	

Model-based Inference under CR

- ▶ Outcome observations are no longer independent in the same cluster
- ▶ Model-based inference

- ▶ linear and generalized linear mixed model - accounting for within-cluster correlation via **random effects** (R-side)

$$y_{kj} = \mu + X_k \delta + b_k + e_{kj}, \quad b_k \sim \mathcal{N}(0, \tau^2), \quad e_{kj} \sim \mathcal{N}(0, \sigma_e^2)$$

- ▶ marginal models - working correlation model in conjunction with **sandwich variance** estimator (G-side)

$$E(y_{kj}) = \mu + X_k \delta, \quad \text{corr}(y_{kj}, y_{kj'}) = \rho$$

- ▶ Fitting via Restricted Maximum Likelihood (REML) and generalized estimating equations (GEE)

- ▶ PROC GLIMMIX in SAS and various packages in R
- ▶ Wald t - test against $H_0: \delta = 0$; **between-within degrees of freedom** ($K - 2$)

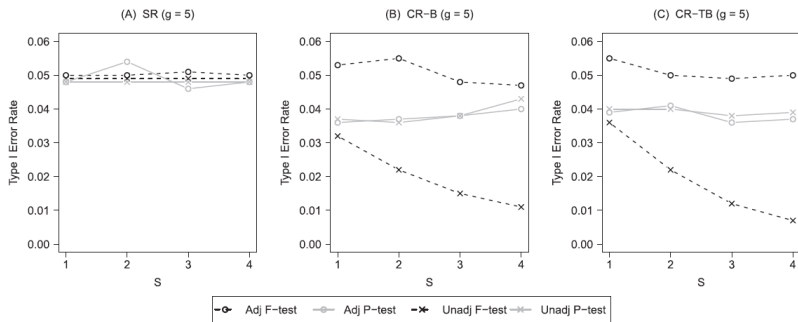
Linear Mixed Models

- ▶ Focus on linear mixed models as the implications for GEE are the same
- ▶ Under CR, the set of cluster-level attributes C_k is balanced by design
 - ▶ X_k is associated with C_k
- ▶ Is **analytical-based adjustment** necessary after **design-based adjustment**?
 - ▶ we assume C_k includes both pure cluster-level attributes & summary statistics for some individual-level characteristics
 - ▶ assume $K \gg p_1$ so that we have enough cluster-level degree of freedom
 - ▶ shall we fit the adjusted linear mixed model (LMM) controlling for C_k , or S_{kj} if C_k contains summary statistics for components of S_{kj} (guesses? - any implications for test size?)

Unadjusted v.s. Adjusted Linear Mixed Models

(Li et al., 2016 Stat Med)

- ▶ CR is still randomization, and therefore $\hat{\delta}$ from the unadjusted LMM is still unbiased
- ▶ But if C_k is associated with y_{kj} (prognostic), the uncertainty of $\hat{\delta}$ may **decrease** as C_p becomes tighter



Unadjusted v.s. Adjusted Linear Mixed Models

(Li et al., 2016, 2017 Stat Med)

- ▶ Unadjusted model-based analysis is **conservative** with more prognostic covariates balanced by design
 - ▶ the model-based standard error **is only consistent under SR** (randomization unconditionally on $\mathbf{C}_k$)
 - ▶ under CR, the model-based standard error is consistent if the model includes $\mathbf{C}_k$
 - ▶ if components of $\mathbf{C}_k$ correspond to components of $\mathbf{S}_{kj}$, then recommend adjusting for $\mathbf{S}_{kj}$ instead
- ▶ If $\mathbf{C}_k$ is not prognostic, then model-based analysis is always valid
 - ▶ but CR can **reduce power loss in SR** due to over-adjustment for $\mathbf{C}_k$
- ▶ A full asymptotic theory for any regression estimator under constrained randomization is developed in Wang and Li¹

¹Wang B, Li F. Asymptotic inference with flexible covariate adjustment under rerandomization and stratified rerandomization. arXiv preprint arXiv:2406.02834. 2024 Jun 5.

Randomization or Permutation Inference

- ▶ Randomization or permutation tests is a robust alternative to model-based methods in CRTs
- ▶ Particularly attractive for CR since the randomness is driven by the randomization space
 - ▶ outcome data are first analyzed based on the actual observed group allocation
 - ▶ the observed test statistic is then referenced against the **exact permutational distribution** calculated from **all other possible allocation schemes**
- ▶ Under the strong null hypothesis of no treatment effect ($H_0: \delta$), the type I error rate of the permutation test will **not exceed the nominal level α**
 - ▶ proved in BIS 537

Efficient Test Statistic

- ▶ Choice of test statistics does not affect **test validity**, but can affect power
- ▶ Consider model-assisted permutation test (Braun and Feng, JASA 2001)
 - ▶ denote $\mathbf{y}_k = (y_{k1}, \dots, y_{k,n_k})'$, under a LMM or GLMM, the **most power permutation test** statistic is $\prod_{k=1}^K f(\mathbf{y}_k)$
 - ▶ for example, the unadjusted LMM corresponds to

$$\begin{aligned} f(\mathbf{y}_k) &= \int (2\pi\sigma_e^2)^{-n_k/2} \exp\left\{-\frac{1}{2\sigma_e^2} \sum_{j=1}^{n_k} (y_{kj} - \mu - X_k\delta)^2\right\} \\ &\quad \times (2\pi\tau^2)^{-1/2} \exp\left\{-\frac{1}{2\tau^2} b^2\right\} db \\ &\propto \exp\left(-\frac{1}{2\sigma_e^2} \left\{ \sum_{j=1}^{n_k} (y_{kj} - \mu - X_k\delta)^2 - \tau^2 W_k \left[\sum_{j=1}^{n_k} (y_{kj} - \mu - X_k\delta) \right]^2 \right\}\right) \end{aligned}$$

where $W_k = 1/(\sigma_e^2 + n_k\tau^2)$

Efficient Test Statistic - Cont'd

- ▶ The exponent in $f(\mathbf{y}_k)$ can be written into

$$\text{const} + \frac{X_k \delta}{\sigma_e^2} (1 - n_k \tau^2 W_k) \sum_{j=1}^{n_k} (y_{kj} - \mu)$$

where ‘const’ stands for terms that are **invariant to permutation** of cluster-level treatment

- ▶ If we parameterize the model such that $X_k \in \{-1, 1\}$, this leaves the overall statistic

$$\sum_{k=1}^K X_k W_k \sum_{j=1}^{n_k} (y_{kj} - \mu)$$

or the covariate-adjusted version

$$\sum_{k=1}^K X_k W_k \sum_{j=1}^{n_k} (y_{kj} - \mu - \mathbf{Z}'_{kj} \boldsymbol{\theta})$$

- ▶ Test statistic does not even depend on the value of δ - uniformly most powerful permutation test (UMPP)
 - ▶ connection to **Neyman-Pearson Lemma & UMP**
- ▶ UMPP test statistic has a clean interpretation
 - ▶ weighted sum of cluster errors (**precision weights** under random intercept model)
- ▶ To operationalize UMPP under CR:
 - ▶ fit the LMM leaving out X_k to obtain W_i , μ or θ (for the adjusted test)
 - ▶ compute test statistics under observed allocation
 - ▶ compute test statistics under all allocations within in the **constrained randomization space**
 - ▶ p-value as the proportions of test statistics under permutation that are as or more extreme than the observed value

Permutation Test - Cont'd

- ▶ The above UMPP is computationally efficient, as it only fits the LMM once and under H_0 (unlike other tests requiring many model fits)
 - ▶ essentially a score permutation test
- ▶ The permutation distribution has to be computed under CR if the design considers CR
 - ▶ if the permutation test instead operates under SR (say the analysts **forgets about the design constraints in the analysis stage**) when CR is used
 - ▶ unadjusted permutation test becomes **conservative**
 - ▶ adjusted permutation test becomes **anti-conservative**
- ▶ UMPP always maintains the type I error rate under CR even without adjusting for C_k (**validity**)
- ▶ UMPP is as powerful as LMM Wald test if LMM is correct, but has larger power if not (**robustness**)

Caveats with Permutation Test

- ▶ We need **at least 20 allocation schemes** to admit a permutation with $\alpha = 0.05$
- ▶ If 10 clusters are equally randomized to 2 arms, we have 252 allocation schemes under simple randomization
- ▶ Constraining to the smallest 19 schemes make the 0.05-level permutation test infeasible (why?)
 - ▶ another caveat with **overly constrained randomization**
- ▶ Confidence interval requires inverting the permutation test and is notoriously computationally difficult
 - ▶ more recent developments on stochastic search algorithm (Rabideau and Wang, Biostatistics 2020)

Does CR helps with inference?

- ▶ For both model-based and randomization-based tests
 - ▶ CR stands for **design-based adjustment**
 - ▶ covariate adjustment in LMM or permutation test stands for **analysis-based adjustment**
- ▶ With prognostic covariates
 - ▶ **Design-based adjustment** and **analysis-based adjustment** alone can improve efficiency for estimating δ the test power
 - ▶ **Design-based adjustment** can further improve power after **analysis-based adjustment**
 - ▶ But often (and as expected) **analysis-based adjustment** provides larger gain in power

References

Ivers, N. M., Halperin, I. J., Barnsley, J., Grimshaw, J. M., Shah, B. R., Tu, K., ... Zwarenstein, M. (2012). Allocation techniques for balance at baseline in cluster randomized trials: a methodological review. *Trials*, 13(1), 1-9.

Dickinson, L. M., Beaty, B., Fox, C., Pace, W., Dickinson, W. P., Emsermann, C., Kempe, A. (2015). Pragmatic cluster randomized trials using covariate constrained randomization: a method for practice-based research networks (PBRNs). *The Journal of the American Board of Family Medicine*, 28(5), 663-672.

Klar N, Donner A (1997). The merits of matching in community intervention trials: A cautionary tale. *Stat Med*.

Raab GM, Butcher I (2001). Balance in cluster randomized trials. *Stat Med*

Moulton, L. H. (2004). Covariate-based constrained randomization of group-randomized trials. *Clinical Trials*, 1(3), 297-305.

References

- Li F, Turner E, Heagerty PJ, Murray DM, Vollmer W, DeLong ER (2017). An evaluation of constrained randomization for the design and analysis of group-randomized trials with binary outcomes. *Stat Med*
- Li F, Lokhnygina Y, Murray DM, Heagerty PJ, DeLong ER (2016). An evaluation of constrained randomization for the design and analysis of group-randomized trials. *Stat Med*
- Braun, T. M., Feng, Z. (2001). Optimal permutation tests for the analysis of group randomized trials. *Journal of the American Statistical Association*, 96(456), 1424-1432.
- Rabideau, D. J., Wang, R. (2020). Randomization-based confidence intervals for cluster randomized trials. *Biostatistics*.
- Zhou, Y., Turner, E. L., Simmons, R. A., Li, F. (2022). Constrained randomization and statistical inference for multiarm parallel cluster randomized controlled trials. *Statistics in Medicine*, 41(10), 1862-1883.